

Design specification: yield-aware grating coupler on 220 nm SOI

Implementation spec for the FDTD loop, GDS, and MPW. 13 September 2026.

This is the engineering document. Physics and claims live in the technical report. An engineer should be able to build the next three artefacts from these pages: a 2-D FDTD ranking check, a focusing-coupler GDS, and a 25 mm² MPW reticle of paired loopbacks.

Goals

Beat the AIM PDK TE vertical coupler (~2.8 dB) by ≥ 0.20 dB in paired on-wafer loopbacks. Hold $P(\text{IL}(1550 \text{ nm}) \leq 2.5 \text{ dB}) \geq 0.80$ in the process model, then confirm empirically. 1 dB bandwidth $\geq 25 \text{ nm}$, on-chip reflection $< -12 \text{ dB}$. Stay inside 150 nm min CD, single shallow etch, no back-reflector unless the PDK already has an overlay.

Non-goals

Sub-0.5 dB on this stack; quantum, cryogenic, 256-channel, TCO, or dual-use write-ups; claiming validation before FDTD and MPW exist.

Key decisions

Decision	Choice	Why
Device	Focusing GC, TE, 1550 nm, 8°	Wafer-probeable; PDK baseline exists
Stack	220 nm / 2 μm BOX / 70 nm etch	Public MPW
Min CD	150 nm	193 nm DUV, conservative
Objective	$E[\text{IL}] + 0.35 \text{ CVaR}_{0.9}$	Chance constraint without fake convexity
Spec	2.5 dB @ 80% yield	Above model mean 2.42 dB; below PDK
v0 geometry	Linear fill apodization	Already computed; ship if FDTD agrees
v1 geometry	Filtered density, $r_{\min} = 75 \text{ nm}$	Adjoint inverse design
Experiment	One MPW, paired loopbacks	91 devices/arm powers 50 vs 70% yield

Locked FDTD stack

Si $n = 3.476$ at 1550 nm, 220 nm device, 70 nm partial etch (150 nm remaining), 2.00 μm BOX $n = 1.444$, Si handle. Fiber: Gaussian, 10.4 μm $1/e^2$ intensity diameter, 8° from normal in the cladding, TE. Offset along z is free (~3.6 μm from the first tooth in the analytic apodized design).

FDTD protocol (stop conditions)

Grid 8 nm max, 4 nm in the grating. Forward: strip mode 20 μm before the taper. Fiber monitor 12 μm wide, 1.5 μm above the cladding, 8° tilt, SMF-28 overlap. Also record on-chip reflection and substrate power.

Sweep order: (1) uniform $f = 0.5$, $\Lambda = 622 \text{ nm}$ — expect ~2.7 dB; if off by $> 0.3 \text{ dB}$, recalibrate η_{2D} before anything else. (2) Analytic apodization — expect ~2.3 dB. (3) Fiber offset $\pm 4 \mu\text{m}$, angle $\pm 2^\circ$. (4) Etch $\pm 10 \text{ nm}$, $t_{\text{Si}} \pm 6 \text{ nm}$. (5) Adjoint loop 150–300 iterations with β continuation and a 20-point process batch every 20 iterations for CVaR. (6) 3-D of the winner only. If 2-D adjoint cannot beat the analytic apodization by 0.15 dB, ship the apodization. Inverse design

is not obligatory.

GDS cell GC_TE_C_8DEG

Focusing grating, 25 lines, 14 μm width; linear taper 500 nm $\rightarrow$ 14 μm over 150 μm ; loopback GC–strip–GC at 127 μm fiber-array pitch; period ladder 600–640 nm step 5 nm; PDK TE GC loopbacks paired on the same die ± 1 mm. PCM: CD SEM 310/310 nm, unpatterned Si thickness, 1 mm spiral.

PR plan

PR1 analytic kit (this repo). PR2 GDS emitter (gdstk). PR3 FDTD driver (Tidy3D or Meep). PR4 tapeout deck (100+100 loopbacks). PR5 measurement notebook (paired t, Clopper–Pearson). Details and open questions (which MPW, poly-Si overlay, O-band vs C-band) are in design/design-spec.md. The MPW statistical protocol is design/mpw-protocol.md.

Risks

Risk	Mitigation
2-D model optimistic vs 3-D	Recalibrate before inverse design; 3-D the winner
Etch-depth mean shift	Period ladder; three periods around 622 nm
Inverse design overfits the simulator	Process batch in the loop; min-CD filter
Claiming 0.5 dB anyway	Do not

MPW in one page

Primary hypothesis: paired same-die mean IL of (PDK – candidate) ≥ 0.20 dB, 95% CI excludes 0. Secondary: Clopper–Pearson interval on $P(\text{IL}_{\text{cand}} \leq 2.5 \text{ dB})$ lies above 0.70; median 1 dB BW ≥ 25 nm. Experimental unit is a loopback, not a wafer. 100 PDK + 100 candidate loopbacks, period ladder, PCM, on one AIM 25 mm² passive MPW (member list price \$28,600 for 20 chips). Analysis locked before data: paired t on H1, Clopper–Pearson on H2. Do not pre-register 0.3 dB or 70% yield at 0.5 dB — that hypothesis is already false in simulation. Full protocol: design/mpw-protocol.md.